Text Feature Engineering Assignment Report

1. Introduction

In this project, a text processing pipeline was developed to convert real-world user-generated reviews into numerical features suitable for machine learning models. The main objective was to apply different text feature engineering techniques such as One Hot Encoding (OHE), Bag of Words (BoW), and TF-IDF, and analyze their effectiveness in representing textual data.

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2. Dataset Collection

A dataset of product reviews was collected from Flipkart using Selenium-based web scraping. The reviews were extracted from a laptop product page and stored in a CSV file with a column named review_text.

- Total reviews collected: ~100
- Data format: CSV
- Source: Flipkart product reviews

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3. Text Preprocessing

Raw text data is often noisy and needs to be cleaned before analysis. The following preprocessing steps were applied:

- Converted all text to lowercase
- Removed punctuation and special characters
- Removed numbers and mixed alphanumeric tokens
- Tokenized text into words
- Removed stopwords (e.g., "the", "is", "and")
- Removed very short and irrelevant words

Example:

Before preprocessing:
"This Laptop is AMAZING!!! Battery is not good 44k price"

After preprocessing:
"laptop amazing battery good price"

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4. Vocabulary Creation

Vocabulary refers to the set of all unique words present in the dataset after preprocessing.

- Vocabulary was generated using CountVectorizer
- Noise such as numbers and meaningless tokens was removed
- The final vocabulary consisted of meaningful words like:
    - battery
    - performance
    - good
    - bad
    - display

Vocabulary size indicates how many unique words exist in the dataset and directly affects feature dimensions.

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5. Feature Engineering

Three different techniques were used to convert text into numerical features:

5.1 One Hot Encoding (OHE)

- Represents whether a word is present or not (0 or 1)
- Does not consider frequency

Advantage: Simple representation
Disadvantage: Ignores word importance and frequency

5.2 Bag of Words (BoW)

- Counts how many times each word appears in a document

Advantage: Captures word frequency
Disadvantage: Treats all words equally (no importance weighting)

5.3 TF-IDF (Term Frequency - Inverse Document Frequency)

- Assigns importance to words based on frequency and rarity

Key idea:

- Common words → low weight
- Rare but meaningful words → high weight

Advantage: Better representation of important words
Disadvantage: Does not capture semantic meaning

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6. Comparison of Methods

Method	Strength	Weakness
OHE	Simple	No frequency or importance
BoW	Captures frequency	No importance
TF-IDF	Captures importance	No semantic understanding

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7. Sparse Matrix Analysis

All feature representations resulted in sparse matrices.

- Most values in the matrix are zero
- Only a few words appear in each review

Why sparse matrices occur:

Each review contains only a small subset of all possible words in the vocabulary.

Issues with sparse matrices:

- High memory usage
- Inefficient for large-scale systems

- Slower computations

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8. TF-IDF Insights

TF-IDF assigns lower weights to common words such as:

- good
- product
- laptop

These words appear frequently across many reviews and therefore provide less distinguishing information.

Higher weights are assigned to unique words such as:

- excellent
- poor
- overheating

These words help in better classification.

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9. Limitations

Bag of Words:

- Cannot understand meaning
- Treats similar words differently (e.g., "good" vs "excellent")

TF-IDF:

- Ignores word order
- Cannot understand context or semantics
- Still treats words independently

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10. Mini Use Case: Sentiment Classification

A Logistic Regression model was used to classify reviews into:

- Positive (1)
- Negative (0)

Process:

- Text converted using BoW and TF-IDF
- Model trained using training data
- Tested on unseen data

Observation:

- TF-IDF performed better than BoW
- Because it focuses on important words

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11. Conclusion

In this project, a complete text processing pipeline was implemented, starting from data collection to machine learning modeling. Among all techniques, TF-IDF proved to be the most effective representation for capturing important textual features.

The project demonstrates how raw text data can be transformed into meaningful numerical features and used for real-world applications such as sentiment analysis.

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12. Future Improvements

- Use advanced models like Word2Vec or BERT
- Improve preprocessing using stemming/lemmatization
- Use larger datasets for better accuracy
- Apply deep learning models

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13. Summary

✓ Data scraped from real-world source
✓ Text cleaned and preprocessed
✓ Vocabulary created
✓ Features extracted using OHE, BoW, TF-IDF

✓ Model trained using Logistic Regression
✓ Performance compared

🎯 This completes the full text feature engineering pipeline.